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Exploration of AI-ML through citizen science approaches

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DELIVERABLE 2.2 – IMPLEMENTING CITIZEN-IN-THE-LOOP IN AI/ML

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SUMMARY: This report is being submitted in accordance with Deliverable 2.2 within workpackage 2 - Communication and Community Engagement. As a follow-up to the concepts and ideas presented in Deliverable 2.1, we move to empirical case studies. We explore the utility of citizen-in-the-loop efforts in feature selection, machine learning baseline training, machine learning targeted training and machine learning updating. We do this at the hand of simulated citizen-in-the-loop strategies in field tests for a smart household budget survey and field tests for a smart travel survey. We conclude that citizen-in-the-loop involvement has added value in feature selection, targeted training and updating. Based on a field test by DESTATIS we also draw a preliminary conclusion that citizen-in-the-loop may be feasible in survey practice. However, during the preparation of this report, we have not been able to perform user experience qualitative studies with citizens/respondents. The broader feasibility of implementing citizen-in-the-loop remains in part an open research question.

1. INTRODUCTION

Accurate and timely training data are a crucial prerequisite to (supervised) AI/ML methods (see also deliverables of AIML4OS WP5). For some AI/ML applications training data can be collected at relatively low cost. However, often an investment is needed in evaluating, annotating and checking events of interest. Apart from costs, annotations and corrections may not be feasible without the involvement of the data subjects. This is the setting of this deliverable. We consider official statistics AI/ML (re)training data settings where the help of 'citizens' is imperative to obtain accurate training data and to keep training data up to date. As a secondary motivation, especially in the official statistics context, we see added value in involving general populations in terms of transparency, engagement and trustworthiness.

This deliverable is the second of two within the topic of human-in-the-loop. The first deliverable set the terminology, lines of thought and a framework, and it functioned as a stepping-stone to this second deliverable. This deliverable proceeds to implementation. It includes simulations and experiences from real-life studies.

Citizen-in-the-loop strategies are an interplay between a statistical sampling viewpoint and a methodological annotator recruitment viewpoint. The sampling viewpoint aims at efficacy of citizen involvement, i.e. AI/ML performance. The annotator viewpoint considers annotator psychology in terms of burden and competence. Together they lead to a focus on efficiency: how to reach and maintain sufficient performance while not overloading or overburdening citizens.

The main elements in our strategy are a distinction of different citizen involvement phases and a trade-off against in-house annotation by staff. We see four phases for a machine learning model: feature selection, baseline training, targeted training and updating. Feature selection corresponds to the choice of auxiliary information to be included in machine learning models. Baseline training refers to a minimal training to get first predictions. Target

training is fine-tuning of the baseline machine learning model. And updating concerns the revision of the machine learning model over time to account for drift in data and/or concepts. Each requires different citizen involvement. The decision as to which annotation queries may be allocated to citizens is set against the proportion which are to be allocated to in-house staff. A key role in the trade-off is annotator bias, i.e. measurement errors made in the annotations.

In this deliverable, we will answer the following research questions:

RQ1. What role is there for citizens in feature selection?
RQ2. What are optimal sampling strategies for citizens in training ML models?
RQ3. What are optimal sampling strategies for citizens in updating ML models?
RQ4. How willing and able are citizens to assist in citizen-in-the-loop tasks?

To answer the fourth question, empirical evidence is needed. We have only been able to evaluate one such study with the help of DESTATIS. Conclusions are preliminary and this question requires more future work and replication of findings.

To date, literature on citizen-in-the-loop within the official statistics setting is thin. Within the Eurostat-funded project Smart Survey Implementation (SSI, see for background and project deliverables https://cros.ec.europa.eu/dashboard/trusted-smart-surveys), both qualitative user experience studies and large-scale field studies have been performed. These were linked to two ESS-surveys, the Household Budget Survey (HBS) and the Harmonized European Time Use Survey (HETUS). Prior to SSI, Statistics Netherlands had performed two field studies, one linked to the HBS and one linked to the Dutch Travel Survey. See Rodenburg, Schouten and Struminskaya (2025) and Carrière, Struminskaya and Schouten (2026) for evaluation of 'citizen involvement' in the HBS. See Remmerswaal et al (2025) for an analysis of respondent behaviour in the Travel Survey. The conclusion from the various studies is that a substantial proportion of citizens/respondents are willing to perform tasks if they understand the rationale behind them. Furthermore, the ability to adequately perform the tasks depends on whether these conform to other activities they are familiar with and perform online in daily life. Tasks that are 'alien' to citizens may be performed with a high frequency of errors or not at all. The UI-UX (user interface – user experience), therefore, needs to be prepared with great care. We must note that these experiences all come from studies embedded in on-going, repeated surveys.

The deliverable, and more generally this task within WP2, links to other AIML4OS Work Packages. The clearest link is to WP5 on standards. In particular, the total machine learning error framework (Puts, Salgado and Daas 2024) is an important starting point to motivate choices in citizen-in-the-loop strategies. To be concrete, citizen-in-the-loop strategies may improve representation of training data and close the gap between training population and target population.

This deliverable is organized as follows: In Section 2, we explain what annotator recruitment options we see as most feasible. We discuss representation in Section 3, which is the main objective of annotator recruitment. Next, we discuss how we answer the research questions in Section 4. In Section 5, we introduce the two empirical studies and discuss how annotator recruitment was implemented. We present and discuss answer to the research questions in Section 6. We end with a discussion in Section 7.

2. STRATEGIES FOR ANNOTATOR SAMPLING

Before moving to the evaluation of citizen-in-the-loop workloads, we first consider how involvement of citizens may be organized. The context in which citizens are invited, and the introduction and motivation of the tasks will have a big impact on their willingness. Furthermore, the user interface of tools designed to perform tasks may strongly affect both willingness and ability.

In Deliverable 2.1, we identified five options to recruit citizens:

1. Personal networks: Small-scale recruitment of citizens through the personal networks of NSI staff. Citizens may also include NSI colleagues.
2. Non-probability-based platforms: Invitations may be posted on (semi-)commercial platforms and/or citizen science platforms.
3. Probability-based dedicated sampling: An NSI may draw samples from population registers and send out dedicated invitations to the sampled population units to participate in one or more phases.
4. Follow-up of respondents in repeated surveys: Respondents from relevant repeated probability-based surveys are invited to participate in a dedicated follow-up study.
5. Smart surveys: Annotation is embedded in repeated smart surveys where the corresponding (smart) data are collected by default.

Table 2.1: Anticipated fitness -for-purpose per citizen-in-the-loop phase. '+' = well-suited, '+/-' = can be used and '-'= not suited.

Recruitment	Phase			
	Feature selection	Baseline training	Targeted training	Updating
Network	+	-	-	+/-
NPS platform	+/-	+/-	-	+/-
Dedicated survey	-	+	+/-	+/-
Follow-up	+/-	+/-	+	+
Smart survey	-	+	+	+

The fitness-for-purpose of the options varies across the phases. In Table 2.1, we provide the anticipated fitness per phase. Feature selection benefits most from in-depth, qualitative study. This means that relatively a lot of time is asked from a citizen, but that the number of citizens can also be relatively small. The best option is network recruitment and perhaps recruitment through platforms or relevant, repeated surveys. Baseline training is very different and requires both quantity and strong representation. A dedicated survey or a 'smart' survey in which data evaluation and validation by respondents is already implemented seem the best options. Targeted training requires knowledge about past events and context and flexibility to select citizens based on that knowledge. A follow-up survey and a 'smart' diary survey offer such knowledge. A dedicated survey may be an option but is less flexible as background of citizens is unknown at the invitation stage. For updating trained models all options seem viable. Features may be revisited, large-scale samples may be needed when time dynamics are high, but also flexibility in the selection may be a crucial asset. We conjecture that this phase depends most strongly on the characteristics of the case study.

In this deliverable we touch on all citizen involvement phases. However, we devote most of the evaluation to the last two phases, advanced learning and updating.

We must note that the studies involved in this deliverable all are set in the context of smart surveys, with one exception which is a study that has a network set-up. Accordingly, while multiple modes of human-in-the-loop are reported in the present report, preliminary analysis is only provided for those two contexts.

3. OBJECTIVES OF ANNOTATION SAMPLING

Before moving to investigations of the balance between efficacy of machine learning models and citizen workload, we need to be more concrete on the objectives. What is effective annotation sampling?

In citizen-in-the-loop, annotation sampling has two steps: sampling of citizens as annotators and within citizens' sampling of events. The second step implies subsampling of events that yet need classification.

Let us first consider annotator/citizen sampling. Machine learning models are usually evaluated in terms of accuracy. However, accuracy depends on the availability of informative features. The availability of features does not (directly) depend on the annotator. A data set may have perfect representation but only a few available features. Performance could therefore be poor. Vice versa, a data set may be very rich in terms of informative features but have very weak representation. The apparent performance may then be misleading, and predictions may be subject to bias and error. The only safeguard is strong representation. The key objective of annotator sampling, therefore, is representative training data and retraining/updating data while avoiding annotator measurement bias. We need to be more rigorous in what we mean by representative.

In the second step, the subsampling of events to be annotated, the citizen workload comes into play. Here, the trade-off is made between representation and burden. In Section 4, we will explain the strategy to implement this trade-off.

In the remainder of this Section, we discuss what we mean by representation and how it translates to citizen-in-the-loop. We note that citizen-in-the-loop recruitment is one option in minimizing total machine learning error as discussed in WP5 of AIML4OS. A detailed account will be given within the final main deliverable.

3.1 Representativeness

The term 'representativeness' has long been used (and still is used) without mathematical rigor. To define representative response/selection, the commonly used approach is to adopt a random response model. To each unit in the target population a response probability is assigned that reflects the selection into the available data, i.e. in the context of this paper the training data. This probability is unknown, except in controlled sampling designs with full response. Its existence has also been the topic of debate, because the probability depends on a wide range of data collection choices. A natural interpretation is that the response probability reflects all variation that is caused by circumstances that the data collector cannot control or is not willing to control.

With the random response model in mind, strong representation is defined as all response probabilities being equal. In terms of missing-data-mechanisms, it means that missing training data is missing-completely-at-random for all possible pairs of estimators/predictors

and population parameters. Strong representation is a conceptual property. It cannot be tested, except again in sampling designs.

A weaker form of representation is needed. It is defined through the employment of auxiliary variables. Imagine a set of auxiliary variables, say X, is available for all units, both responding/available and not responding/available. The response propensity with respect to X, $P(\text{Response}|X = x)$, denoted as ρ_X, is the projection of the response probability on the space formed by the auxiliary variables. Weak representation with respect to X is defined as constant response propensities ρ_X. In words, the selected auxiliary information does not detect any deficiency in representation in training data.

Now, with these definitions in mind, we can (and should) distinguish ML performance/accuracy and representation. Representativeness and accuracy are different properties. Training data can be perfectly representative, i.e. each event or person has the same probability of inclusion in the training data but lead to weak performance. This is for two reasons. The first reason is that representation is not about the availability of features. The available features may not or only be weakly informative about the target variable. The second reason is that representation is a property of bias and not of variance. The size of the training data set may be very small and still have a low feature space resolution. However, ML training tends to be performed in data sets with many features and the feature selection is greedy by nature. Consequently, representation must come before accuracy. The only safeguard against spurious high accuracy is to start from a representative training data set. This is the main motivation for separating training into baseline training, advanced training and updating. Baseline training focusses on representation whereas advanced training and updating shift focus to accuracy.

The weak representation definition is the stepping-stone to indicators for representation. The general approach is to measure variation in propensities. When the focus is estimation or prediction of population means or totals then a frequently used indicator is the coefficient of variation of response propensities, i.e. training data selection propensities. It is the ratio of the standard deviation of response propensities ($S(\rho_X)$) to the average response propensity ($\bar{\rho}_X$), i.e.:

$$CV(\rho_X) = \frac{S(\rho_X)}{\bar{\rho}_X}.$$

3.2 Measuring representativeness

While available indicators do require choice, the true influential decision is in the auxiliary information to be included. The indicators merely transform multi-dimensional evidence on lack of representation to lower, interpretable dimensions. However, with every different choice of X, the values of indicators will change. Obviously, the set and range of available auxiliary variables will be limited. Regardless of choice, there is also a limit to what can be detected.

We may distinguish the following types of variables:

- Target variable Y
- Features completely available outside data collection:
  - Important for ML: X_1
  - Ultimately not important for ML: X_2

- Features not available outside data collection:
  - Important for ML and observed: Z_1
  - Important for ML but not observed: Z_2

In WP5, the following is proposed:

- For training, the auxiliary variables are selected from X_1 and X_2, i.e. we consider $CV(\rho_{X_1,X_2})$
- For updating, the variables are selected from X_1 and Z_1, i.e. we consider $CV(\rho_{X_1,Z_1})$

The features Z_2 are introduced to highlight that representation measurement will, in general, not be without risk of bias. The more features are missing, i.e. the weaker the performance of the ML model, the greater the impact that the response rate may have on actual bias in prediction. In other words, the more risk that new units are different from what has been observed before.

In the introduction of the case studies in Section 5, we will specify the different types of variables that are available and that we will employ.

4. METHODS

To answer the four research questions, we perform three types of analyses:

A. Confrontations of features used by citizens and features selected in machine learning models (RQ1);
B. Simulations of different annotation sampling scenarios to find the balance between workload and efficacy (RQ2 and RQ3);
C. Investigation of actual willingness rates and performance in data collection (RQ4);

To investigate the role of citizens in feature selection, we compare feature importances in machine learning models to information that citizens will employ in annotation. We explain how we do this in Section 4.1.

As explained in Section 2, annotation sampling consists of two steps: citizen sampling and event sampling within citizens. In Section 4.2, we propose simulation strategies that allow us to investigate the second step, event sampling.

Sampled events to be annotated by citizens likely are subject to nonresponse and to measurement error. In Section 4.3, we discuss how we evaluate these errors.

We stress that we have limited empirical evidence to study actual willingness and ability of citizens.

4.1 Feature selection

We investigate the role of features as follows:

1. We estimate machine learning models for different sets of features
2. We rank features by feature importances
3. We categorize features as inside/outside the knowledge of (average) citizens
4. We omit features that are outside the knowledge of citizens and evaluate the impact on accuracy

4.2 Simulating annotation sampling

In evaluating citizen sampling and event sampling, we distinguish the three phases that follow feature selection: baseline training, targeted training and updating. Annotation sampling may be implemented as follows in two sequential steps:

- Decide to recruit citizens with relevant background characteristics based on representativity indicators
- Decide to select an event based on prediction results

The following simulations are conducted to evaluate these two steps:

Baseline training:

- We omit subpopulations based on known background characteristics at the onset of the data collection to study the impact of non-representative citizen sampling
- We omit subsets of events
- We vary the size of the baseline training set by including data in batches

Targeted training

- We omit subpopulations based on known characteristics at the onset
- We omit subsets of events
- We vary targeted training thresholds

Updating

- We offer events in batches chronologically
- We vary retraining thresholds

We calculate a number of metrics per simulation:

- Rate of events that do not meet (re)training thresholds
- Coefficient of variation of training data
- Accuracy unbalanced and balanced overall
- Sensitivity and specificity of individual classification categories
- The distance in classification tree (hierarchical classifications only) for incorrect labels

4.3 Actual willingness and performance

In field studies, the focus shifts to actual behaviour by participating persons and households.

We derive the following metrics:

- Actual annotation rate (including nonresponses by annotators)
- Rate of correct and incorrect adjustments made by respondents
- Relation between correct and incorrect adjustments made by respondents and their background characteristics
- Relation between correct and incorrect adjustments made by respondents and the ML target variable
- Distance in classification tree (hierarchical classifications only) for incorrect labels

5. CASE STUDIES

The results in this paper are based on two case studies:

1. Classification of purchased products/services from texts printed on receipts;
2. Classification of travel purposes from location tracking data;

Case studies have been categorized on four properties in deliverable D2.1: dynamics in time (drift of concepts and features), burden to the annotator (annotator willingness), complexity to the annotator (annotator error) and need for context (feature selection). The first case study was categorized as highly time-dynamic, modest burden, complex because of COICOP definitions, and no need for context. The second case study scores differently: low time dynamics, high burden, low complexity and a need for context as persons cannot interpret GPS coordinates.

5.1 COICOP classification in the Household Budget Survey

For COICOP classification, we report two studies, one in Germany and one in the Netherlands. The German study is based on the 2026 fieldwork of the German HBS that included citizen-in-the-loop options. The Dutch study is based on simulations using a fully labelled set of products in a 2021 experimental study performed within the ESTAT-funded ESSnet project Smart Surveys. We explain the study designs and the evaluations that we perform.

5.1.1 Simulating different annotation scenarios in the Dutch HBS

In 2021, Statistics Netherlands (SN) conducted an experimental study into a smart Household Budget Survey. A simple random sample of 1600 households was selected from the Dutch population register and divided across a range of experimental conditions. Households were invited to install a cross-platform app. All households had the choice between scanning receipts and manual data entry throughout the two-week reporting period. A random subset of households was asked to also check, and if needed correct, optical character recognition and receipt language processing that was performed real-time in the app. Classification of product texts was conducted solely for manual data entry.
Afterwards all receipts were checked and classified by SN staff. Missing and spurious products were added, if not noticed by respondents. Products on scanned receipts were assigned a COICOP category. The labelled products are the starting point for the simulation study.

The following simulations are performed, and the corresponding research questions (RQ) as described in the Introduction are also indicated

- Subset the shops included in the training data (RQ2)
- Gradually offer batches of products purchased in the same month and vary annotation thresholds (RQ2)
- Gradually offer batches of products from new months and vary annotation thresholds (RQ3);
- Compare simulated patterns to actual patterns for unlabeled data (RQ2 and RQ3);
- Compare different product categories (RQ2 and RQ3).

Given that all receipts were handled, we can evaluate the impact of citizen sampling through the shops that they visit and the impact of different thresholds in event sampling.

5.1.2 Evaluating citizen-in-the-loop in the German HBS

We present preliminary results from methodological analyses of the German Household Budget Survey (HBS) of 2026. DESTATIS employs an app developed by Hbits for HBS. The app is available both as web app and as cross-platform mobile app. The app includes a tool

where households can check and correct receipt language processing and COICOP classification. COICOP classification follows three steps:

I. Automatic classification using a range of methods: For details, we refer to Minnen et al. (2025) and Van Hoorde et al. (2026).
II. Citizen-in-the-loop checking and correcting/supplementing: Households are asked to assist in classification. Scanned receipts are collected in a tray (a type of inbox) and are displayed as tentative data. The tentative classifications are given in two options: (i) most likely correct and (ii) not clear and without a suggested category.
III. In-house checks and corrections: Staff are trained to check both respondent-handled and respondent-non-handled receipts. Most importantly, missing classifications needed to be manually added. Furthermore, automatic classifications or human-in-the loop corrections/additions had to be checked in terms of a quick check of data rows.

Through the DESTATIS social media channels, the German population was informed of the option to participate in diary surveys by DESTATIS. The invited households in the 2026 HBS are a quota sample taken from the households that registered themselves as willing to participate. The January and February samples were smaller to test if the app functioned as expected. The samples of subsequent months, starting in March, had a regular sample size.

The following analyses were performed:

- What is the actual checking rate of classifications by households?
- How does this classification rate depend on time in study?
- What type of households (e.g. size, income level, age) actively check/adjust classifications?
- What is the false adjustment rate of households checking classifications?
- What is the false non-adjustment rate of households checking classifications?
- To what extent does the error rate correlate to the uncertainty that comes out of the pipeline itself?
- To what extent does the error rate depend on the product (shop, COICOP)?
  To what extent does the error rate depend on the novelty of the product (name)?

5.2 Travel purpose classification in the Dutch Time Use Surveys

Statistics Netherlands developed a cross-platform smart travel survey app in-house. In 2018 and in 2021, Statistics Netherlands (SN) fielded an experimental smart travel survey. The survey was administered through an app that performed location tracking through a mix of GPS, GSM and Wi-Fi. The 2018 and 2021 apps were different, but both performed real-time stop-track segmentation. However, the 2018 app only allowed for labeling of stops and tracks. The 2021 app offered various options for respondents to edit stops and tracks. They could change begin and end times of stops and tracks, remove stops or tracks, and add stops or tracks. Respondents were asked to provide the travel modes and the purposes of stops.
Using the 2021 app, an additional small-scale study was conducted directed at training of travel mode and travel purpose prediction. A group of ten students employed the app on multiple smartphones and made a range of travels following pre-specified scenarios. The scenarios were anticipated to be especially difficult for stop-track segmentation and for prediction. The students kept detailed diaries.

The following evaluation was done for the 2021 small-scale test (corresponding Research Questions indicated in brackets)

- Confrontation of features for machine learning models and citizens (RQ1)

The following simulations were performed for the 2021 large-scale test:

- Omit different sets of features (RQ1)
- Subset the population of travelers based on age (RQ2 and RQ3)
- Vary the length of the location tracking period (RQ3)

6. RESULTS OF THE EMPIRICAL STUDIES

We describe the three case studies. We like to express our explicit gratitude to DESTATIS who offered insight into their HBS fieldwork free of cost.

6.1 Annotator sampling strategies in the Dutch HBS

We elaborate the five simulations performed on the HBS labelled data.

6.1.1 Subset the shops included in the training data

Shops differ greatly in how they print product texts. Some shops use only strings of 12 character positions whereas others use 20 character positions. Some shops use their own house brand names whereas others do not have house brands. Some shops frequently use abbreviations whereas others do not. The shop name is a relevant feature in product classification, therefore.

In the creation of training/reference data, Statistics Netherlands has three options: scanner transaction data product descriptions, databases of printed texts and manual labeling.

In the simulation, we left out individual chains of shops and tested the impact. Here, we show one example: 'HOOGVLIET'. For this shop only scanner transaction data descriptions were available. All other shops have been included in training.

In each simulation, we performed five runs. In each run, 20 batches of 50 products were sampled and batches were offered one by one to the model. Citizen-in-the-loop annotation was then evaluated for three thresholds, 85, 90 and 95 on the classification similarity score. Under the 85 threshold, citizen-in-the-loop labeling is needed only for scores lower than 85. For the 95 threshold, all scores below 95 are assigned. Five runs were chosen because of computation times. As a consequence, some of the series were not perfectly smooth. Higher numbers of runs would

We compared two event sampling designs: sampling based on sales amounts with replacement and simple random sampling without replacement. The first design mimics what would be observed in real practice once a new shop is included. The second design gives insight into the diversity of product texts.

In Tables 6.1 and 6.2, we display results for sampling based on sales amounts with replacement and simple random sampling without replacement, respectively. In each table, we give the average rates of automatic classification, automatic classification that gives perfect classification, CITL classification along with the accuracy and tree distance of misclassified products.

Table 6.1: Simulations for a shop (HOOGVLIET) with only scanner data product descriptions based on sales amount sampling with replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1,00	0,85	0,19	0,08	0,10	0,43	0,00	0,15	0,81	0,27	0,28	0,89	2,6	2,8	1,8
2	1,00	0,88	0,48	0,07	0,12	0,75	0,00	0,21	0,90	0,14	0,27	0,63	2,9	2,6	2,3
3	1,00	0,89	0,58	0,08	0,15	0,82	0,00	0,22	0,85	0,31	0,25	0,49	2,6	3,0	2,6
4	1,00	0,93	0,61	0,07	0,15	0,82	0,00	0,16	0,88	0,32	0,25	0,49	2,5	2,7	2,5
5	1,00	0,95	0,71	0,08	0,17	0,82	0,00	0,13	0,78	0,20	0,20	0,38	2,7	2,8	2,5
6	1,00	0,95	0,76	0,09	0,20	0,80	0,00	0,16	0,74	0,28	0,22	0,32	2,5	2,7	2,6
7	1,00	0,94	0,77	0,11	0,17	0,79	0,00	0,19	0,74	0,28	0,21	0,33	2,6	2,9	2,5
8	1,00	0,95	0,82	0,10	0,19	0,81	0,00	0,23	0,77	0,29	0,22	0,31	2,5	2,7	2,5
9	1,00	0,94	0,71	0,09	0,17	0,81	0,00	0,19	0,94	0,28	0,20	0,41	2,6	2,8	2,5
10	1,00	0,93	0,77	0,07	0,18	0,81	0,00	0,26	0,80	0,29	0,24	0,34	2,7	2,8	2,5
11	1,00	0,98	0,83	0,08	0,25	0,84	0,00	0,09	0,77	0,31	0,31	0,29	2,5	2,5	2,6
12	0,99	0,95	0,80	0,08	0,23	0,79	0,02	0,23	0,80	0,23	0,23	0,35	2,7	2,6	2,4
13	1,00	0,96	0,86	0,09	0,21	0,81	0,00	0,17	0,68	0,15	0,14	0,23	3,1	3,0	2,7
14	1,00	0,96	0,89	0,09	0,25	0,79	0,00	0,24	0,64	0,17	0,23	0,24	3,0	2,7	2,4
15	1,00	0,97	0,87	0,09	0,19	0,81	0,00	0,22	0,74	0,29	0,17	0,22	2,5	2,9	2,6
16	1,00	0,98	0,86	0,09	0,22	0,81	0,00	0,12	0,78	0,31	0,23	0,32	2,4	2,5	2,3
17	1,00	0,97	0,87	0,09	0,21	0,83	0,00	0,16	0,80	0,27	0,25	0,27	2,6	2,6	2,4
18	1,00	0,98	0,86	0,09	0,22	0,81	0,00	0,11	0,90	0,29	0,25	0,26	2,6	2,6	2,5
19	0,99	0,96	0,86	0,11	0,22	0,80	0,07	0,20	0,73	0,28	0,29	0,32	2,7	2,5	2,2
20	1,00	0,96	0,89	0,12	0,26	0,84	0,00	0,24	0,68	0,33	0,16	0,29	2,4	2,8	2,3

Table 6.2: Simulations for a shop (HOOGVLIET) with only scanner data product descriptions based on simple random sampling without replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1,00	0,87	0,27	0,04	0,04	0,14	0,00	0,13	0,73	0,21	0,18	0,77	3,5	4,1	3,7
2	1,00	0,84	0,21	0,03	0,04	0,15	0,00	0,16	0,79	0,20	0,23	0,79	3,2	4,3	4,4
3	1,00	0,81	0,20	0,06	0,07	0,29	0,00	0,19	0,80	0,22	0,20	0,81	3,3	4,0	4,0
4	1,00	0,84	0,21	0,03	0,04	0,13	0,00	0,16	0,79	0,20	0,17	0,81	3,4	4,1	4,3
5	0,99	0,85	0,20	0,02	0,02	0,13	0,01	0,15	0,80	0,17	0,25	0,82	3,6	3,8	3,4
6	1,00	0,83	0,27	0,01	0,02	0,08	0,00	0,17	0,73	0,21	0,19	0,75	3,3	4,1	4,3
7	0,99	0,86	0,21	0,03	0,03	0,13	0,01	0,14	0,79	0,16	0,22	0,80	3,4	4,2	4,1
8	1,00	0,81	0,21	0,02	0,02	0,12	0,00	0,19	0,79	0,19	0,20	0,81	3,5	4,3	3,8
9	0,99	0,81	0,22	0,01	0,02	0,09	0,01	0,19	0,78	0,18	0,19	0,80	3,5	4,1	3,6
10	1,00	0,84	0,21	0,01	0,02	0,12	0,00	0,16	0,79	0,21	0,21	0,79	3,3	4,3	3,9
11	1,00	0,79	0,20	0,01	0,01	0,03	0,00	0,21	0,80	0,23	0,16	0,81	3,2	4,1	3,8
12	1,00	0,87	0,23	0,03	0,05	0,25	0,00	0,13	0,77	0,15	0,23	0,77	3,5	4,2	4,4
13	0,99	0,82	0,25	0,03	0,05	0,17	0,01	0,18	0,75	0,17	0,20	0,77	3,7	3,9	4,3
14	1,00	0,85	0,25	0,02	0,02	0,11	0,00	0,15	0,75	0,24	0,19	0,77	3,2	4,2	4,2
15	1,00	0,87	0,24	0,03	0,03	0,13	0,00	0,13	0,76	0,17	0,18	0,76	3,6	4,1	4,1
16	1,00	0,85	0,22	0,05	0,06	0,23	0,00	0,15	0,78	0,23	0,17	0,80	3,2	4,0	3,9
17	1,00	0,85	0,23	0,04	0,05	0,17	0,00	0,15	0,77	0,20	0,16	0,77	3,4	4,4	3,9
18	1,00	0,85	0,24	0,02	0,04	0,14	0,00	0,15	0,76	0,18	0,15	0,77	3,5	4,3	4,1
19	0,99	0,86	0,23	0,04	0,05	0,20	0,01	0,14	0,77	0,17	0,16	0,78	3,7	4,3	4,1
20	1,00	0,88	0,21	0,03	0,03	0,13	0,00	0,12	0,79	0,23	0,21	0,79	3,3	4,0	4,0

The following findings are typical for shops where only scanner data are available:

- Performance is very poor and a sizeable amount of citizen-in-the-loop involvement is needed to get to acceptable levels;
- Tree distances show that misclassifications can be severe;
- Sampling without replacement demonstrates that a large number of products need to be seen first before accuracy increases. A series of 1000 unique products (20 batches of 50 products) is still relatively small;

The more general conclusion is that baseline training is needed for all individual shops. Targeted training is not suited for entering entire new shops.

6.1.2 Gradually offer batches of products purchased in the same month and vary annotation thresholds

In the second simulation, we involve shops for which we have printed receipt texts linked to COICOP. We now isolate samples of their products and pretend they have not yet been seen. We then offer batches iteratively and monitor automatic classification, citizen-in-the-loop classification, accuracy and tree distances of misclassified products.

In Tables 6.3 and 6.4, we present results for a typical shop ('PLUS') where printed texts are available. Results are based on five independent runs of 20 batches of 50 products. We performed sampling with replacement based on sales amounts (Table 6.3) and simple random sampling without replacement (Table 6.4).

Findings are very different when compared to shops with only scanner data descriptions:

- Performance is much higher and CITL assignment rates can be much lower;
- When considering samples with replacement then the CITL rate decreases already after small numbers of products;

These findings tell us that targeted training may be applied to shops that have already been observed at a certain rate through baseline training.

Table 6.3: Simulations for a shop (PLUS) with labeled receipt texts based on sales amount sampling with replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1	0,85	0,00	0	0,00	0,00	0	0,15	1,00	0,64	0,65	1	1,3	0,7	0
2	1	0,85	0,39	0	0,00	0,96	0	0,15	0,61	0,63	0,78	1	1,4	0,7	0
3	1	0,85	0,52	0	0,02	0,94	0	0,15	0,48	0,56	0,79	1	1,6	0,7	0
4	1	0,89	0,57	0	0,05	0,91	0	0,11	0,43	0,59	0,78	1	1,5	0,6	0
5	1	0,95	0,63	0	0,08	0,90	0	0,05	0,37	0,68	0,79	1	1,1	0,6	0
6	1	0,93	0,73	0	0,05	0,92	0	0,07	0,27	0,67	0,79	1	1,1	0,6	0
7	1	0,91	0,61	0	0,04	0,92	0	0,09	0,39	0,59	0,80	1	1,4	0,7	0
8	1	0,92	0,71	0	0,09	0,93	0	0,08	0,29	0,65	0,77	1	1,3	0,5	0
9	1	0,93	0,69	0	0,09	0,86	0	0,07	0,31	0,62	0,83	1	1,4	0,6	0
10	1	0,94	0,73	0	0,09	0,89	0	0,06	0,27	0,65	0,79	1	1,2	0,5	0
11	1	0,96	0,71	0	0,10	0,85	0	0,04	0,29	0,66	0,81	1	1,2	0,6	0
12	1	0,94	0,65	0	0,06	0,93	0	0,06	0,35	0,65	0,80	1	1,2	0,6	0

Table 6.4: Simulations for a shop (PLUS) with labeled receipt texts based on simple random sampling without replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1	0,85	0,00	0	0,00	0,00	0	0,15	1,00	0,64	0,78	1,00	1,3	0,7	0,0
2	1	0,81	0,05	0	0,00	0,00	0	0,19	0,95	0,53	0,72	1,00	1,7	1,0	0,0
3	1	0,74	0,08	0	0,00	0,14	0	0,26	0,92	0,52	0,75	1,00	1,7	0,9	0,0
4	1	0,81	0,07	0	0,00	0,00	0	0,19	0,93	0,52	0,76	1,00	1,7	0,8	0,0
5	1	0,79	0,09	0	0,00	0,05	0	0,21	0,91	0,47	0,73	0,99	1,9	1,0	0,1
6	1	0,81	0,14	0	0,00	0,13	0	0,19	0,86	0,54	0,76	1,00	1,6	0,7	0,0
7	1	0,86	0,20	0	0,00	0,06	0	0,14	0,80	0,58	0,76	0,99	1,6	0,9	0,3
8	1	0,84	0,15	0	0,00	0,05	0	0,16	0,85	0,52	0,72	0,99	1,7	1,0	0,2
9	1	0,89	0,15	0	0,00	0,07	0	0,11	0,85	0,59	0,74	0,98	1,4	0,9	0,4
10	1	0,82	0,14	0	0,01	0,26	0	0,18	0,86	0,48	0,71	0,98	1,8	1,0	0,6
11	1	0,78	0,21	0	0,02	0,27	0	0,22	0,79	0,53	0,73	0,97	1,7	1,0	0,5
12	1	0,83	0,13	0	0,00	0,30	0	0,17	0,87	0,55	0,73	0,99	1,5	0,9	0,1

6.1.3 Gradually offer batches of products from new months and vary annotation thresholds

Receipt texts change over time. New products are launched. Some products change texts. And other products disappear. The dynamics in the receipt texts is high. Some shops introduce up to 500 new texts per month. Here, we simulate time dynamics by sampling texts from a new month. We do not change the classification model.

In Table 6.5, we present the results of simulations when offering batches of 50 products from a new month. In the first batches, most of the texts are assigned to citizen-in-the-loop annotation. However, after five batches the assignment stabilizes to around a quarter of the receipts. Results for other shops are comparable.

Table 6.5: Simulations for a shop (PLUS) with labeled receipt texts based on simple random sampling without replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

Batch	Average auto	Average exact	Average label	Accuracy	Tree distance
1	0,00	0,00	1,00	1,00	0,0
2	0,33	0,94	0,67	0,99	0,08
3	0,52	0,59	0,48	0,93	0,34
4	0,75	0,62	0,25	0,92	0,24
5	0,67	0,62	0,33	0,95	0,21
6	0,75	0,69	0,25	0,91	0,27
7	0,75	0,64	0,25	0,92	0,23
8	0,79	0,71	0,21	0,93	0,21
9	0,79	0,67	0,21	0,94	0,17
10	0,77	0,74	0,23	0,95	0,20
11	0,85	0,69	0,15	0,89	0,32
12	0,75	0,71	0,25	0,96	0,13

We conclude:

- New months can severely distort classification performance;

- Citizen-in-the-loop annotation is important in updating classification at the beginning of new months;

6.1.4 Compare simulated patterns to actual patterns for unlabeled data

To this point, we have simulated patterns based on fully labelled data. Here, we compare actual observed patterns to simulated patterns. For this purpose, we use the 2021 smart Household Budget Survey field test.

The 2021 field test consisted of a simple random sample of households from the population register. One person within each household was sampled and personally invited to act as proxy. The population sample was linked to a range of demographic and socio-economic background variables, including household income and household size. The linkage to a wide set of background variables allows for an assessment of representation. Rodenburg, Schouten and Struminskaya (2025) employs R-indicators[1] and coefficients of variation. They estimate indicators for households that install and register the mobile app, for households that become active and for households that complete the study. They estimate unconditional and conditional partial R-indicators at the variable-level[2] to assess the impact of individual variables. Table 6.6 is taken from Rodenburg et al. (2025). They conclude that response propensity variation is sizeable. Educational level of the reference person has the strongest impact, followed by household income and age of the reference person. These results point us at a potential over/underrepresentation of certain shops and certain products.

Table 6.6. R indicators, CV's, unconditional partial R-indicators (Pu) and conditional partial R-indicators (Pc) for household characteristics for the 2021 population sample. 95% confidence intervals are given for R and CV (source: Rodenburg et al. 2025).

	Registered				Active			
	R	CV	Pu	Pc	R	CV	Pu	Pc
Netherlands	0.77 [0.71, 0.79]	0.56 [0.44, 0.68]			0.78 [0.73, 0.81]	0.64 [0.50, 0.78]		
Age			0.06	0.02			0.05	0.02
Household size			0.03	0.00			0.02	0.01
Origin			0.03	0.01			0.04	0.02
Education level			0.09	0.05			0.09	0.04
Urbanization			0.00	0.00			0.01	0.00
Homeownership			0.04	0.01			0.04	0.01
Income			0.07	0.03			0.07	0.03

[1] R-indicators are short for representativity indicators. These indicators evaluate the distances between respondents and non-respondents for auxiliary variables. Loosely formulated they correspond to the numerator in the coefficient of variation.

[2] Partial R-indicatros decompose the variance in estimated response propensities. Unconditional partial R-indicators give the contribution by a variable unadjusted for collinearity with other variables. Conditional partial R-indicators give the unique contribution. Values closer to zero indicate less impact.

In Table 6.7, we show the results for the classification of products from the field test. Since these products were not labelled by households, we cannot evaluate performance of the classification. We can, however, evaluate the impact on the classification threshold on the proportion of products that would have been allocated to households had we asked them to. We added 12 batches of 250 products each. The products were ordered chronologically as they were submitted by households.

The field test results show us that citizen-in-the-loop sampling would have allocation rates that are very similar to what we observed in simulated data. Around a quarter of products is assigned under the strictest 95 threshold. Hence, the potential impact on shop and product representation is not seen in these proportions. However, in-depth analysis of receipts and receipt texts would be needed to see if indeed performance is worse for certain groups of products or certain stores.

Table 6.7: Classification of 2021 field test data in 12 batches of chronologically ordered products.. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household.

Batch	Average automatic			Average exact			Average labeled			Average label rate		
	85	90	95	85	90	95	85	90	95	85	90	95
1	250	241	171	124	124	124	0	9	79	0,00	0,03	0,32
2	250	239	176	123	123	131	0	11	74	0,00	0,04	0,30
3	250	238	187	125	126	135	0	12	63	0,00	0,05	0,25
4	249	238	194	128	129	139	1	12	56	0,01	0,05	0,23
5	248	240	183	126	127	141	2	10	67	0,01	0,04	0,27
6	249	239	189	123	126	138	1	11	61	0,00	0,04	0,25
7	248	242	192	118	122	140	2	8	58	0,01	0,03	0,23
8	249	242	194	128	129	143	1	8	56	0,00	0,03	0,22
9	249	240	184	124	124	138	1	10	66	0,00	0,04	0,27
10	250	243	191	124	125	140	0	7	59	0,00	0,03	0,23
11	250	240	197	124	127	140	0	10	53	0,00	0,04	0,21
12	249	244	196	131	136	149	1	6	54	0,00	0,03	0,22

6.1.5 Compare different product categories

As a last evaluation, we look at specific product categories. We considered two groups of products, one where we expected a lot of subtle variation in texts and one where we expected less diversity. We chose dairy products (01.1.4) as a diverse group of products and fruits and vegetables (01.1.6 and 01.1.7) as a less diverse group. Fruits and vegetables have diverse product names, but we expect only few variants per type of fruit or type of vegetable. For dairy products we expect a lot of variants.

Tables 6.8 and 6.9 give the simulation results for dairy products and fruits and vegetables, respectively. Contrary to our expectation, we find relatively similar automatic and citizen-in-the-loop classification rate patterns for the two groups of products. Also, accuracy patterns are comparable for the two groups of products. The average CITL allocation rates are relatively high for the first few batches and then stabilize. The rates differ but little from the rates for all products.

Hence, our first explorations do not point at a strong impact of the type of products on classification rates and classification success. This is good news. We conjecture that also the impact of less representative training/updating data with respect to household characteristics will be less influential.

Table 6.8: Simulations for dairy products for a shop (PLUS) with labeled receipt texts based on simple random sampling without replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1,00	0,78	0,12	0,0	0,0	0,0	0,00	0,22	0,88	0,11	0,29	0,90	3,4	3,1	2,4
2	1,00	0,88	0,17	0,0	0,0	0,0	0,00	0,12	0,83	0,14	0,17	0,86	3,2	3,2	2,6
3	1,00	0,82	0,17	0,0	0,0	0,0	0,00	0,18	0,83	0,14	0,27	0,85	3,1	3,1	3,3
4	1,00	0,85	0,14	0,0	0,0	0,0	0,00	0,15	0,86	0,12	0,22	0,89	3,3	3,2	2,9
5	1,00	0,78	0,17	0,0	0,0	0,0	0,00	0,22	0,83	0,19	0,31	0,86	3,0	2,6	2,4
6	1,00	0,75	0,16	0,0	0,0	0,1	0,00	0,25	0,84	0,11	0,31	0,86	3,5	3,4	3,1
7	1,00	0,85	0,15	0,0	0,0	0,0	0,00	0,15	0,85	0,11	0,19	0,86	3,2	3,3	3,4
8	1,00	0,85	0,23	0,0	0,0	0,1	0,00	0,15	0,77	0,16	0,23	0,83	3,2	3,0	1,8
9	1,00	0,79	0,20	0,0	0,0	0,1	0,00	0,21	0,80	0,15	0,27	0,86	3,4	3,3	2,2
10	1,00	0,87	0,21	0,0	0,0	0,2	0,00	0,13	0,79	0,13	0,25	0,85	3,2	2,9	2,2
11	1,00	0,85	0,21	0,0	0,0	0,1	0,00	0,15	0,79	0,18	0,28	0,83	3,0	2,9	2,4
12	1,00	0,85	0,17	0,0	0,0	0,1	0,00	0,15	0,83	0,13	0,25	0,85	3,3	3,1	2,6
13	1,00	0,89	0,28	0,0	0,0	0,1	0,00	0,11	0,72	0,20	0,25	0,81	3,1	3,1	1,8
14	1,00	0,89	0,25	0,0	0,0	0,1	0,00	0,11	0,75	0,17	0,19	0,77	3,2	3,3	2,7
15	1,00	0,89	0,22	0,0	0,0	0,2	0,00	0,11	0,78	0,20	0,25	0,82	3,0	2,8	1,9
16	1,00	0,91	0,31	0,0	0,0	0,2	0,00	0,09	0,69	0,11	0,17	0,77	3,3	3,0	2,1
17	1,00	0,85	0,26	0,0	0,0	0,2	0,00	0,15	0,74	0,17	0,28	0,81	3,1	2,8	2,5
18	1,00	0,82	0,25	0,0	0,0	0,2	0,00	0,18	0,75	0,17	0,28	0,81	3,1	3,0	2,1
19	1,00	0,90	0,31	0,0	0,0	0,2	0,00	0,10	0,69	0,11	0,19	0,77	3,3	3,0	1,8
20	1,00	0,89	0,27	0,0	0,0	0,2	0,00	0,11	0,73	0,18	0,23	0,81	3,2	3,0	2,0

Table 6.9: Simulations for fruits and vegetables for a shop (PLUS) with labeled receipt texts based on simple random sampling without replacement. Average auto = average proportion of products that are classified automatically, average exact = average proportion of exact matches, average label = average proportion of products that is given to a responding household, accuracy = average proportion correct, tree distance = distance in tree for incorrect classifications.

	Average auto			Average exact			Average label			Accuracy			Tree distance		
	85	90	95	85	90	95	85	90	95	85	90	95	85	90	95
1	1,00	0,93	0,07	0,00	0,00	0,00	0,00	0,07	0,93	0,16	0,17	0,94	3,2	3,3	2,3
2	1,00	0,89	0,07	0,00	0,00	0,13	0,00	0,11	0,93	0,22	0,18	0,94	2,9	3,2	3,4
3	1,00	0,89	0,10	0,00	0,00	0,43	0,00	0,11	0,90	0,21	0,19	0,93	2,8	3,0	2,0
4	1,00	0,97	0,09	0,00	0,00	0,26	0,00	0,03	0,91	0,21	0,20	0,93	2,9	3,0	1,6
5	1,00	0,89	0,09	0,00	0,00	0,30	0,00	0,11	0,91	0,21	0,18	0,91	2,8	3,2	3,2
6	1,00	0,84	0,16	0,00	0,02	0,50	0,00	0,16	0,84	0,17	0,26	0,90	2,9	3,0	1,8
7	1,00	0,96	0,23	0,00	0,01	0,53	0,00	0,04	0,77	0,24	0,17	0,81	2,7	3,0	2,2
8	1,00	0,89	0,17	0,00	0,00	0,46	0,00	0,11	0,83	0,16	0,13	0,86	3,1	3,4	2,3
9	1,00	0,93	0,16	0,00	0,01	0,70	0,00	0,07	0,84	0,23	0,21	0,88	2,9	3,1	2,5
10	1,00	0,90	0,22	0,00	0,02	0,81	0,00	0,10	0,78	0,20	0,16	0,81	2,8	3,1	2,4
11	1,00	0,93	0,17	0,00	0,01	0,53	0,00	0,07	0,83	0,19	0,13	0,85	3,0	3,3	2,3
12	1,00	0,92	0,26	0,00	0,02	0,64	0,00	0,08	0,74	0,19	0,15	0,79	3,1	3,2	2,4
13	1,00	0,90	0,25	0,00	0,01	0,48	0,00	0,10	0,75	0,18	0,17	0,80	3,1	3,3	2,2

14	1,00	0,95	0,25	0,00	0,01	0,63	0,00	0,05	0,75	0,22	0,13	0,82	2,8	3,1	2,0
15	1,00	0,91	0,21	0,00	0,01	0,44	0,00	0,09	0,79	0,18	0,18	0,81	3,0	3,2	2,4
16	1,00	0,90	0,21	0,00	0,03	0,40	0,00	0,10	0,79	0,21	0,17	0,84	2,8	3,0	2,4
17	1,00	0,92	0,28	0,00	0,02	0,54	0,00	0,08	0,72	0,16	0,19	0,76	3,0	3,0	2,2
18	1,00	0,95	0,24	0,00	0,00	0,73	0,00	0,05	0,76	0,15	0,12	0,80	3,3	3,5	2,1
19	1,00	0,91	0,25	0,00	0,02	0,63	0,00	0,09	0,75	0,23	0,20	0,79	2,9	3,1	2,4
20	1,00	0,93	0,30	0,00	0,01	0,70	0,00	0,07	0,70	0,17	0,15	0,73	3,2	3,3	2,4

6.2 Fieldwork experiences in the 2026 German HBS

Before we answer the analysis questions, we note that results are preliminary and based on three months of fieldwork data in 2026. The first two months were relatively small. Furthermore, the data collection comprises only of specific months of the year (quarter 1) which may have an impact on the type of products that are reported. A total of 470 households submitted around 80k products in the first three months via the receipt scanning functionality within the app. The findings are based on these households and products.

DESTATIS was asked the following questions:

- What is the actual checking rate of classifications by households?
  For around 2/3 of the products, there was an automatically predicted COICOP category. Only a very small proportion of these classifications was corrected by respondents. Respondents did supplement most of the missing classifications. Around 1/6 of the missing classifications remained unlabelled, i.e. 1/18 of the total products.
- How does this classification rate depend on time in study?
  There is decrease in the proportion of classifications that was corrected and that was supplemented during the one-month reporting period. The exception are larger uncorrected and non-labelled products at the very last day of data collection. This is however an artefact as receipts with relatively more missing classifications tended to remain in the inbox of respondents.
- What type of households (e.g. size, income level, age) actively check/adjust classifications?
  Editing behaviour was compared for household income quintiles, household size and household type. There is some preliminary evidence that lower income groups and single parents less often supplement missing classifications. However, the sample is still too small for strong conclusions.
- What is the false adjustment rate of households checking classifications?
  The false adjustment encompasses two types of actions: a false correction of an automatically predicted classification and an incorrect label of a missing classification. Since the proportion of automatic classifications that was adjusted is very small, false adjustment is mostly about incorrect supplements. Post-survey staff checks indicate that the proportion of incorrect labels is very small. However, at the time of writing it is unclear to what extent staff members actively checked respondent-labelled codes. Preliminary results point at weak relations between the type of household and the false adjustment rate.
- What is the false non-adjustment rate of households checking classifications?
  Checks by staff indicate that false non-adjustment rates are very small. However, it is suspected that staff concentrated efforts at missing classifications and did not

focus extensively on checking the automatic predictions nor the adjustments by households in detail. Hence, the observed false non-adjustment rate may be incorrect.

- To what extent does the error rate correlate to the uncertainty that comes out of the pipeline itself?
  There is no clear association between the certainty rate that comes out of the automatic prediction and the adjustment rate. Neither is there a clear association to false adjustments.
- To what extent does the error rate depend on the product (shop, COICOP)?
  Figure 6.1 displays the tree distance in adjustments that staff made relative to what respondents labelled. A 'level *k*' switch means that the classification was wrong at the *k*-th digit. The numbers observed are small. Results are preliminary and do not allow for strong conclusions
- To what extent does the error rate depend on the novelty of the product (name)?
  This question could not yet be answered at the time of writing.

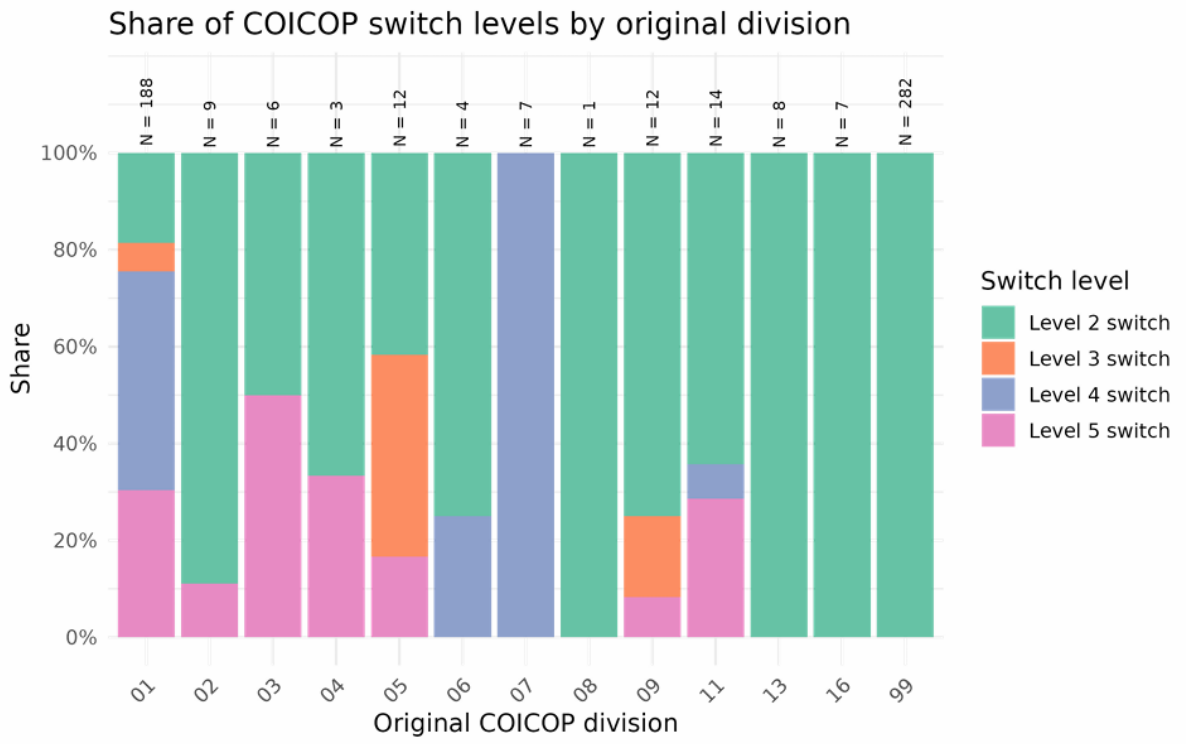


Figure 6.1: Overview of the distance of switches in classification between staff and household per main COICOP category.

We draw the following conclusions from the German HBS:

- Households are willing to perform citizen-in-the-loop actions in a system that is designed with strong user interaction in mind. We must also note, however, that the sample was drawn from a self-selection larger sample.
- Households are mostly supplementing labels and not adjusting labels. It is unclear whether they should have made more adjustments at this point.
- There is no indication that willingness deteriorates during the reporting period.

6.2 Annotator sampling in the Dutch Travel Survey

We move to another survey and context, that of time use. The results presented in this section are linked to a travel survey, which is a specific type of time use survey. We anticipate that results can be generalized to more general time use surveys such as the Harmonized European Time Use Survey (HETUS).

We analyze two data collections: a small-scale dedicated study with detailed diary keeping and a large-scale field test under the flag of the Dutch Travel Survey.

6.3.1 Feature selection in the small-scale student 2021 test

The small-scale study was focused on feature selection in the classification of travel modes and the classification of travel purposes. Detailed results have been reported in Zahroh et al. (2025) and Klingwort et al. (2025 and 2026). Here, for the sake of illustration we focus on travel purpose.

A broad set of features was linked from points-of-interest data in Open Street Maps (OSM). For different radii (20, 50 200m) it was determined whether a list of object types was present such as schools, shops, sport centres, public transport, etc.

Different machine learning models were selected and applied. The best performing model was extreme gradient boosting. Table 6.10 contains the selected features including relative feature importances.

We tried to mimic what information a respondent would use when asked to label stops. Stops can be both intermediate during travels as well as the end of travels. We omitted different subsets of features that we think respondents would not use, we omitted very short stops that would be harder for respondents to classify and we omitted home stops that would be very easy for respondents to classify. The features that we omitted about tracks referred to the duration/distance/speed/transport modes of travels preceding a stop. For the locations, we omitted features on preceding and succeeding stops and the number of visits to the location. In Table 6.11, we show the different sets of features and stops that we evaluated. Around 900 stops were shorter than 10 minutes. Around 2000 stops were home stops.

Table 6.11: Different instances of model fitting where either feature sets or stop types were omitted.

Model label	Population	Features	Days	#records
Full	Full	All	All	4961
NoLoc	Full	Without features about location	All	4961
NoTrack	Full	Without features about tracks	All	4961
NoLocTrack	Full	Without features location or tracks	All	4961
NoShort	Stops >10 minutes	All	All	4026
NoHome	No home stops	All	All	2928

Table 6.12 : Model performance and most important features for the models in Table 6.11.

Labell	Accuracy (unbal)	Top 5 features
Full	77	duration, #visits, n25_shops, time-of-day, prop visits same label

NoLoc	74	duration, n25_house, n25_shops, time-of-day, travelled distance
NoTrack	78	duration, #visits, n25_shops, time-of-day, prop visits same label
NoLocTrack	74	duration, n25_house, n25_shops, time-of-day, n200_routes
NoShort	78	duration, #visits, time-of-day, n25_shops, prop visits same label
NoHome	71	duration, n25_shops, next stop is work, time-of-day, age

In Tables 6.12 and 6.13, we summarize the models. The impact of omitting features is relatively modest. The duration of the stop is the strongest feature. Omitting the second largest feature, the number of visits to the location, does lead to a drop in performance. This is mostly for the home stops. This means that for home stops machine learning models and respondents would use different types of information. It also means that performance of machine learning models will depend on the length of the reporting period, whereas this would be irrelevant for respondents. The omission of short stops is influential for classification of picking up/bringing persons or goods (transport) and for transit stops. Here, we expect that respondents will have difficulties. Leaving out the home stops, we notice an improvement in the classification of visits and work.

We conclude that machine learning and respondents may supplement each other as they are likely to involve different information and one may compensate for the weaknesses of the other.

Table 6.13: Sensitivity and specificity of travel purpose categories for the different subsets of features and stops.

Label	transport		school		other		transit		sports		home		visit		work		shop	
	se	sp	se	sp	se	sp	se	sp	se	sp	se	sp	se	sp	se	sp	se	sp
Full	61	97	68	99	42	97	63	99	49	99	95	91	40	97	79	96	77	96
NoLoc	57	97	65	100	32	97	67	99	50	99	93	83	21	98	78	96	77	95
NoTrack	62	97	63	100	40	96	66	99	52	99	94	91	46	98	81	97	79	96
NoLocTrack	57	97	65	100	32	97	65	99	53	95	92	82	15	98	79	96	78	96
NoShort	53	98	73	99	37	96	38	100	46	99	96	87	45	97	77	96	75	97
NoHome	66	94	67	99	44	93	66	99	57	99	-	-	58	95	86	93	81	92

6.3.2 Simulating annotator sampling in the large-scale 2021 test

Respondents in the large-scale smart travel survey were invited to provide labels for all of their stops and tracks, For the travel purposes, i.e. the labels of stops, they could choose from a list of nine categories: home, school/study, work/office, bringing/picking up goods or persons, shopping, sports, family/friends visit, transit and other. For the travel modes, i.e. the labels of tracks, they had the following options: car as driver, car as passenger, motorcycle, bus, tram, train, subway, bicycle and on-foot. -Some of the respondents had the option to fully edit the stop-track segmentation. In the analysis, however, we pooled all labelled stops and tracks without distinguishing the experimental condition. As for the small-scale pilot described in Section 6.3.1, we linked point-of-interest data from Open Street Maps (OSM).

The 2021 field test was preceded by a field test in 2018. The 2018 field test was evaluated in detail on its representation by Lugtig, Roth and Schouten (2022). Figure 6.2 is taken from their evaluation and plots the R-indicators of registration, activity and completion against registration, activity and completion rates. The diagonal lines correspond to constant values of the coefficient of variation of estimated response propensities. The authors conclude that there is substantial variation in response propensities. They concluded that variables showing the strongest contribution to response propensity variation are household income and age. Background variables on ownership of different means of transport (car, motorcycle, moped)

were linked as well but gave relatively little contribution to variation. From these findings, we deduce that training data may be subject to lack of representation. This may be especially true for travel purpose prediction,

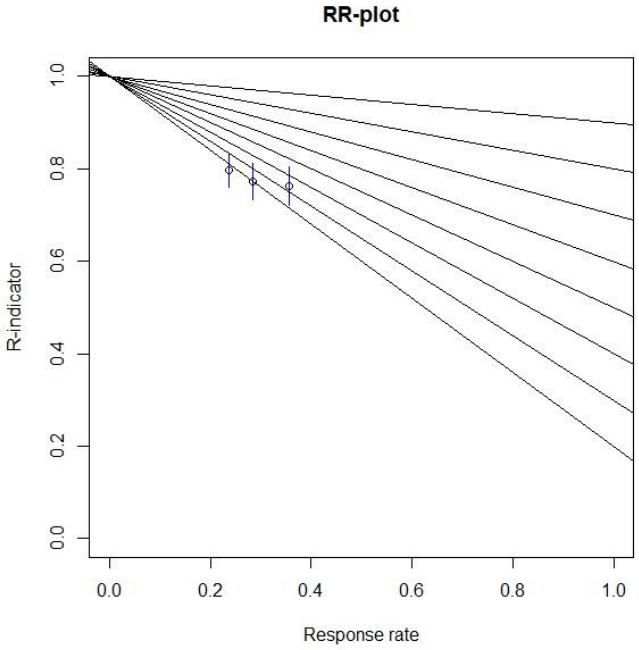


Figure 6.2: Response-representativity (RR) plot for three stages of data collection: app registration, at least one day of tracking and a complete diary. Diagonal lines represent coefficients of variation (CV). Source: Lugtig, Roth and Schouten, 2022.

Table 6.14: Properties of extreme gradient boosting models fitted for different subpopulations and different tracking periods.

Model label	Population	Features	Days	#records
Full	Full	All	All	4961
25+	25+ years	All	All	4252
55-	55- years	All	All	3214
7d	Full	All	7	4854
6d	Full	All	6	4511
5d	Full	All	5	4035
4d	Full	All	4	3579
3d	Full	All	3	2973
2d	Full	All	2	2392

Table 6.15 : Model performance and most important features for the models in Table 6.13.

Labell	Accuracy (unbal)	Top 5 features
Full	77	duration, #visits, n25_shops, time-of-day, prop visits same label
25+	77	duration, #visits, n25_shops, time-of-day, prop visits same label
55-	76	duration, #visits, n25_shops, n25_house, prop visits same label
7d	76	duration, #visits to location at t, n25_shops, n25_house, time-of-day
6d	78	duration, n25_house, n25_shops, time of day, #visits to location at t
5d	75	duration, n25_house, n25_shops, time of day, #visits to location at t

4d	75	duration, n25_house, n25_shops, time of day, #visits to location at t
3d	74	duration, n25_house, n25_shops, time-of-day, travelled distance
2d	74	duration, n25_house, n25_shops, time-of-day, travel duration

Table 6.16: Sensitivity and specificity of travel purpose categories for the different subpopulations and tracking periods.

Label	Transport		school		other		transit		sports		home		visit		work		shop	
	se	sp	se	sp	se	sp	se	sp	se	sp	Se	sp	se	sp	se	sp	se	sp
Full	61	97	68	99	42	97	63	99	49	99	95	91	40	97	79	96	77	96
25+	65	96	40	99	41	96	53	99	48	99	94	91	40	97	80	96	78	96
55-	67	97	70	99	36	96	64	99	44	99	94	91	41	98	78	96	77	96
7d	61	97	64	100	58	96	64	99	48	99	94	89	35	97	77	96	77	96
6d	62	97	65	100	38	96	59	99	44	99	94	89	34	98	78	96	79	95
5d	62	96	63	100	40	96	55	99	43	99	94	89	32	98	77	96	77	96
4d	60	97	58	99	36	96	53	99	43	99	94	89	30	98	75	96	79	95
3d	58	97	47	99	37	95	48	99	43	99	95	88	29	98	73	96	78	95
2d	58	97	31	100	33	95	54	99	40	99	95	86	24	98	70	96	77	95

In the simulations, we looked at the age class of respondents and at the duration of location tracking, because age is a strong predictor of travel frequency, travel mode and travel purpose. We omitted young respondents and we omitted older respondents. We varied the reporting period from two days to the full seven days. We give the results in Tables 6.14 and 6.15. We conclude that omitting age groups had a minor impact on predictions. Age is selected as a feature but with a low to modest feature importance. Hence, the potential under- or overrepresentation of age groups does not have a strong impact. The tracking period length is an influential feature. Pe[3]rformance gradually deteriorates when shortening the tracking period. This is especially true for school, work and sports. As longer tracking periods are more demanding for respondents, this leads to a trade-off between citizen-in-the-loop sampling and performance. Citizen-in-the-loop sampling will have a larger added value for the most-visited destinations. A way to deal with this is to ask at the onset of the study whether the respondent will go to school, work and/or sports during the upcoming week.

7. DISCUSSION

In this deliverable, we set out to find empirical evidence that citizen-in-the-loop approaches may be beneficial in the application of machine learning models in official statistics setting. We had to do so based on simulation studies in which fully labelled data is available. Consequently, we can present relatively little support for the actual feasibility of citizen-in-the-loop in survey practice. The exception is a study by DESTATIS where households are actively involved in classification or product categories. This study shows that implementation is feasible.

We conducted the evaluation at the hand of four research questions: The role of citizens in feature selection, the role of citizens in training, the role of citizens in (re)training/updating and the actual willingness and ability of citizens.

3

We looked at feature selection in one study, travel mode and travel purpose prediction. We found evidence that information/features important in machine learning are different from those used by respondents. We also learned that machine learning and citizen-in-the-loop may be complementary in that respondents may be strong at classifying events that are harder for machine learning and vice versa.

In the simulation studies connected to training and retraining, we focused on representation and the trade-off of representation against performance. We found that citizen-in-the-loop involvement is especially useful in targeted training and in updating. Baseline training is too demanding and will lead to too many requests. However, once a model has had baseline training, there are still specific events where respondent/citizen annotation may be very powerful. In the case studies this holds for new/rarely visited shops, new products and locations that are visited multiple times but with little context information. We also found that in real-life studies response propensities may have strong variation across relevant population subgroups. For baseline training such imperfect representation may be devastating. However, for targeting and updating the emphasis shifts to maintaining representation of features not available in samples but available during baseline training.

All in all, we conclude that there is a clear case for citizen-in-the-loop to improve representation of machine learning from a theoretical perspective.

Moving back to the respondent/person side of citizen-in-the-loop. The 2026 German Household Budget Survey gives evidence of citizens being willing to help in improving predictions. There is also some evidence that they do a relatively good job. Evidence is thin, however, and more is needed to make firm statements about citizen-in-the-loop in practice. This will remain a topic for future study and replication.

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